

Covariance: $Cov(X, Y) = E[XY] - E[X][Y]$

$Cov(aX, bY) = abCov(X, Y)$ $Cov(X_1 + X_2, Y) = Cov(X_1, Y) + Cov(X_2, Y)$

Correlation: $Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$ $Corr(aX + b, cY + d) = Corr(X, Y)$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$$

$$\prod_{i=1}^p \frac{1}{a_i} = \sum_{i=1}^p \frac{c_i}{a_i}$$

Taylor's Formula: $f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$

Stationary: MA 天然Stationary, AR $1 - \phi B$ 没有根在单位圆上 1. $\mu = E(Y_t) = c$ 2. $\gamma(k) = Cov(Y_t, Y_{t+k})$

only depends on k

Causality: AR 特征方程的根在单位圆外

Invertibility MA 特征方程的根在单位圆外

Inverse Matrix 若 $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, 则 $A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, $|A| = ad - bc$.

$\gamma(k) : MA(q) : Y_t = \theta_0 Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$

$\gamma(k) = \sigma^2 \sum_{i=0}^{q-k} \theta_i \theta_{i+k}$, $k = 0, 1, \dots, q$; $\gamma(k) = 0$, $k > q$.

$Z \sim N(0, \sigma^2)$, $E(Z^2) = \sigma^2$ and $E(Z^4) = 3\sigma^4$

ARMA 课程定义: $Y_t - \phi_1 Y_{t-1} - \dots - \phi_p Y_{t-p} = Z_t - \theta_1 Z_{t-1} - \dots - \theta_q Z_{t-q}$

ARMA R 定义: $Y_t - \phi_1 Y_{t-1} - \dots - \phi_p Y_{t-p} = Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$

Measure of Serial Dependence II - Partial Autocorrelation

Definition 6. The **lag k Partial Autocorrelation function (PACF)** of a weakly stationary time series $\{X_t\}$ is defined by

$$\phi_{kk} := \text{Corr}(X_t, X_{t+k} | X_{t+1}, \dots, X_{t+k-1}).$$

Evaluation of PACF and its estimation

Theorem 2. Let $\{X_t\}$ be weakly stationary with ACF $\{\rho(h)\}$. Then the lag k PACF ϕ_{kk} could be solved through

$$\begin{pmatrix} \phi_{k1} \\ \vdots \\ \phi_{kk} \end{pmatrix} = \begin{pmatrix} \rho(0) & \cdots & \rho(k-1) \\ \vdots & \ddots & \vdots \\ \rho(k-1) & \cdots & \rho(0) \end{pmatrix}^{-1} \begin{pmatrix} \rho(1) \\ \vdots \\ \rho(k) \end{pmatrix}$$

Decomposition of time series:

$$X_t = \underbrace{T_t}_{(\text{Trend})} + \underbrace{S_t}_{(\text{Seasonality})} + \underbrace{N_t}_{(\text{Noise})}$$

$T_t + S_t \Rightarrow$ Macroscopic Component $N_t \Rightarrow$ Microscopic Component

Estimation of trends without seasonality: 3种方法 1. Least Squares Method 2. Filtering 3. Differencing

1. Least Squares Method: 记矩阵 X 的转置是 X'

$$\text{由于 } t = 1, 2, 3, \dots, n \text{ 我们记 } X = \begin{pmatrix} 1 & 1 & \cdots & 1^k \\ 1 & 2 & \cdots & 2^k \\ \vdots & \vdots & & \vdots \\ 1 & n & \cdots & n^k \end{pmatrix} Y = (X_1, X_2, \dots, X_n)' \quad X_t \text{ 是 } t \text{ 时刻的观测值}$$

Goal is to minimize $RSS = \sum_{t=1}^n (X_t - T_t)^2 = (Y - X\beta)'(Y - X\beta) \Rightarrow \hat{\beta} = (X'X)^{-1}X'Y$

2. Filtering

Filter 种类:

1. Low-pass filter: Filter out high-frequencies (volatile) signals Retains low-frequencies (smooth) signals

2. High-pass filter: Filter out low-frequencies (smooth) signals Retains high-frequencies (volatile) signals

Filter Example:

1. Moving Average Filter

$S_t = \frac{1}{2q+1} \sum_{r=-q}^q X_{t+r}$, where S_t is the filtered value at time t , and $2q + 1$ is the window width.

2. Spencer 15-point Filter

$S_t = \sum_{r=-7}^7 w_r X_{t+r}$, where the weights w_r are specifically defined as
 $w = \frac{1}{320} \times [-3, -6, -5, 3, 21, 46, 67, 74, 67, 46, 21, 3, -5, -6, -3]$.

Theorem:

k 阶多项式 $T_t = c_0 + c_1 t + \dots + c_k t^k$ 可以无失真地(Passes through unchanged)通过滤波器

$\hat{T}_t = \sum_{r=-s}^s a_r T_{t+r}$ 的充要条件是滤波器权重 a_r 满足以下两个条件: 1. $\sum_{r=-s}^s a_r = 1$ 2. $\sum_{r=-s}^s r^j a_r = 0$, 对所有 $j = 1, 2, \dots, k$

3. Differencing

概念:

一阶差分 First order: $\Delta X_t = X_t - X_{t-1}$ 二阶差分 Second Order: $\Delta^2 X_t = \Delta(\Delta X_t)$

Backshift operator B : $BX_t = X_{t-1}$, $B^k X_t = X_{t-k}$

Differencing operator Δ : $\Delta X_t = (1 - B)X_t$ $\Delta^k X_t = (1 - B)^k X_t$

用法: Differencing removes trend:

If $X_t = \alpha + \beta t$ 一阶差分后: $\Delta X_t = X_t - X_{t-1} = \alpha + \beta t - [\alpha + \beta(t-1)] = \beta$ 线性趋势被完全去除

If $X_t = \gamma t^p$ 一阶差分后: $\Delta X_t = X_t - X_{t-1} = \gamma t^p - \gamma(t-1)^p = p\gamma t^{p-1} - C_2^p \gamma t^{p-2} + \dots$ 次数降低的多项式

In general $X_t = T_t + N_t$ 其中trend 部分 $T_t = \sum_{j=0}^p a_j t^j$ p 阶差分后: $\Delta^p X_t = p! a_p + \Delta^p N_t$

Estimating/Removing seasonal effect 1. Moving average method 2. Seasonal differencing

Seasonal cycles: General Decomposition $\Rightarrow X_t = T_t + S_t + N_t$ Multiplicative Seasonal Component

$\Rightarrow X_t = T_t S_t N_t$

Seasonal component S_t : period = d 1. $S_{t+d} = S_t$ 2. $\sum_{j=1}^d S_j = 0$

Difficulties when both T_t and S_t exist, need to separate the effect of trend T_t and seasonal effect S_t

1. Moving average method 基于 $X_t = T_t + S_t + N_t$

Step 1: estimate the trend 移动平均滤波器必须具有长度 $d \Rightarrow \sum_{j=1}^d S_j = 0$

1. 当 $d = 2q + 1$: $\hat{T}_t = \frac{1}{d} \sum_{r=-q}^q X_{t+r}$ 2. 当 $d = 2q$: $\hat{T}_t = \frac{1}{d} (\frac{1}{2} X_{t-q} + X_{t-q+1} + \dots + X_{t+q-1} + \frac{1}{2} X_{t+q})$

Step 2: estimate the seasonal components $\Rightarrow$ 给定从 $t = 1, \dots, nd$ 的样本

1. 计算去趋势序列: $D_t = X_t - \hat{T}_t$ 2. 计算去趋势序列的平均值: $\bar{D} = \frac{1}{nd} \sum_{t=1}^{nd} D_t$ 3. 估计季节性成分

$\hat{S}_j = \frac{1}{n} \sum_{k=0}^{n-1} (D_{kd+j} - \bar{D})$

Step 3: Use any filter for the series $X_t - \hat{S}_t$ to get an improved $\tilde{T}_t$ (can set $\tilde{T}_t = T_t$ if you are satisfied with the filter in Step 1.)

$X_t = \tilde{T}_t + \hat{S}_t + \hat{N}_t$

2. Seasonal differencing

Seasonal differencing: $\Delta^{(d)} X_t = (1 - B^d) X_t = X_t - X_{t-d}$

Noticed that we have $\Delta^{(d)} X_t = S_t - S_{t-d} + N_t - N_{t-d} = N_t - N_{t-d}$ 只是消除影响不是分离估计

After decomposition or differencing:

Check the residual: 分解法得到的残差: $\hat{N}_t = X_t - \tilde{T}_t - \hat{S}_t$

差分法得到的残差: $\hat{N}_t = X_t - X_{t-d}$

Stochastic process

$X_t : t = 1, 2, \dots, n$ $X_1, X_2, \dots, X_n$

随机性来源: $\Omega = (\omega_1, \omega_2, \dots)$, 每个 ω 会产生一个特定的效果 $\omega_i \rightarrow X_t(\omega_i)$

对于一个特定的 ω 我们会产生一系列的 $X_t(\omega)$ 我们称为 **a sample function/ realization/ sample path** 对于某个固定的 t $X_t(\cdot)$ 是一个随机变量

实际上我们只能观察到一条样本路径, 即使用某一个 ω 得到的

Finite dimensional distribution function

$F_{\mathbf{t}}(\mathbf{x}) = P(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n), \mathbf{x} = (x_1, x_2, \dots, x_n), \mathbf{t} = (t_1, \dots, t_n)$

Strictly stationary: distribution of a process does not change over time

A process $\{X_t\}$ is said to be strictly stationary if: for all $n, (t_1, t_2, \dots, t_n)$ and h we have

$$(X_{t_1}, \dots, X_{t_n}) \stackrel{d}{=} (X_{t_1+h}, \dots, X_{t_n+h})$$

" $\stackrel{d}{=}$ " means "equal in distribution", i.e.,

$$F_{\mathbf{t}}(\mathbf{x}) = P(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n) = P(X_{t_1+h} \leq x_1, \dots, X_{t_n+h} \leq x_n) = F_{\mathbf{t}+h}(\mathbf{x})$$

Weakly stationary (second order stationary/wide-sense stationary)

If $E(X_t) = \mu$ and $\text{Cov}(X_t, X_{t+h}) = \gamma(h)$ 仅与 h 有关 $\Rightarrow \{X_t\}$ 是弱平稳的

Autocovariance function & Autocorrelation function

ACVF: $\gamma(h) = \text{Cov}(X_t, X_{t+h})$, 描述了随机过程在相隔 h 个时间单位的两个观测值之间的协方差

ACF: $\rho(h) = \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} = \frac{\gamma(h)}{\gamma(0)}$ 在 $[-1, 1]$, 衡量相隔 h 个时间单位的观测值之间的线性相关程度

Under stationarity, we have

1. $\gamma(0) = \text{Var}(X_t) = \text{Var}(X_{t+h})$ 在任何时间点的方差都相同

2. $\gamma(-h) = \gamma(h)$ 向前看 h 步和向后看 h 步的协方差相同

3. $\rho(-h) = \rho(h)$ 向前看 h 步和向后看 h 步的相关性相同

我们注意到他们都是 **population quantities**, 它涉及到随机过程的具体分布, 实际中我们要使用样本估计

Sample ACVF: $C_h = \frac{1}{n} \sum_{i=1}^{n-h} (X_i - \bar{X})(X_{i+h} - \bar{X})$

Sample ACF: $r_h = \frac{C_h}{C_0}$ C_0 是样本方差 (是 n 不是 $n-1$)

为什么 **ACVF** 除以 n ?

Let $\tilde{X}_i = X_i - \bar{X}$ be the centered observations.

With $C_h = \frac{1}{n} \sum_{i=1}^{n-h} \tilde{X}_i \tilde{X}_{i+h}$, the sample covariance matrix can be expressed as:

$$\widehat{\text{Var}}(\mathbf{X}) = (C_{|i-j|})_{i,j=1}^n = \frac{1}{n} \mathbf{M} \mathbf{M}'$$

where $\mathbf{M}$ is a matrix constructed from the centered observations.

For any vector $\mathbf{a}$:

$$\mathbf{a}' \widehat{\text{Var}}(\mathbf{X}) \mathbf{a} = \frac{1}{n} (\mathbf{a}' \mathbf{M}) (\mathbf{M}' \mathbf{a}) = \frac{1}{n} \|\mathbf{M}' \mathbf{a}\|^2 \geq 0$$

proving non-negative definiteness.

Using $\frac{1}{n-h}$ would introduce different scaling factors for different lags, breaking the quadratic form structure and potentially violating non-negative definiteness.

Correlogram - ACF plot

只有当滞后 h 不超过样本量 n 的三分之一时 ($h \leq n/3$), 样本自相关 r_h 才是可靠的。

$$r_0 = \frac{C_0}{C_0} = 1$$

我们常常关心 $\text{Corr}(X_t, X_{t+h}) = 0 \Leftrightarrow r_h$ significantly difference from 0?

If X_t is white noise, $r_h \sim N(0, \frac{1}{n})$, 在 ACF 图上绘制水平线在 $\pm \frac{2}{\sqrt{n}}$, 超出就认为显著不为 0

Short memory vs long memory: Short Memory = Short term correlation Long Memory = long term correlation

Notice:

1. 我们可以用 ACF 图来识别时间序列模式, 但是如果数据中存在趋势那么 X_t 变为非平稳序列, 直接看 ACF 不可

靠，需要先去除趋势

2. 对于 $Y_t = Y_{t-1} + Z_t$ 和 $Y_t = at + Z_t$, ACF 衰减的很慢, 这表明他们是非平稳的, 应当对这些数据做 **detrending** 和 **filtering**

Notations

ARMA(p, q) model is the most common model for **stationary** time series

$$\underbrace{Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p}}_{\text{Autoregressive (AR)}} = \underbrace{Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q}}_{\text{Moving Average (MA)}}$$

Y_t is observation, $Z_t \sim \text{WN}(0, \sigma^2)$ is white noise.

Also it could be write as $\phi(B)Y_t = \theta(B)Z_t$

$$\phi(B) = I - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p,$$

$$\theta(B) = I - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q,$$

$\phi(B)$ and $\theta(B)$ are **characteristic polynomials** without common roots, i.e., no x s.t. $\phi(x) = \theta(x) = 0$

ARIMA(p, d, q) model is the most common model for **non-stationary** time series is

$$\phi(B) \underbrace{(1 - B)^d}_{\text{Integrated}} Y_t = \theta(B) Z_t.$$

If Y_t follows an **ARIMA(p, d, q)** model, then $\triangle^d Y_t = (1 - B)^d Y_t$ follows an **ARMA(p, q)** model, where $(1 - B)^d Y_t$ is the d-times differenced series from Y_t .

$$(1 - B)^d Y_t = \sum_{k=0}^d \binom{d}{k} (-B)^k Y_t = \sum_{k=0}^d \binom{d}{k} (-1)^k Y_{t-k}.$$

Moving Average Model

A stochastic process $\{Y_t\}_{t=1,2,\dots}$ follows an **MA(q)** model if

$$Y_t = Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q} \text{ where } Z_t \sim \text{WN}(0, \sigma^2).$$

MA(q) is a **stationary process**

Proof:

$$\mathbb{E}(Y_t) = 0$$

$$\text{Var}(Y_t) = (1 + \theta_1^2 + \cdots + \theta_q^2) \sigma^2$$

$$\gamma_k = \text{Cov}(Y_t, Y_{t+k}) = \begin{cases} 0 & , \quad |k| > q \\ \sigma^2 \sum_{i=0}^{q-|k|} \theta_i \theta_{i+|k|} & , \quad |k| \leq q, (\theta_0 = -1) \end{cases}$$

Invertibility

Definition: 如果白噪声 Z_t 可以表示为过去观测值 $Y_t, Y_{t-1}, \dots$ 的线性组合, 则该模型是可逆的 (invertible)。

i.e. $Z_t = \sum_{k=0}^{\infty} \psi_k Y_{t-k}$ 其中 $\sum_{k=0}^{\infty} |\psi_k| < \infty$

An **MA(q)** process Y_t is invertible if the roots of the equation $\theta(B) = 0$ all lie outside the unit circle

Proof:

$$Y_t = Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q}$$

$$= \theta(B) Z_t$$

$$= (1 - \bar{\theta}_1 B)(1 - \bar{\theta}_2 B) \cdots (1 - \bar{\theta}_q B) Z_t$$

$\frac{1}{\bar{\theta}_k}$ s are the roots of the equation $\theta(B) = 0$

$(1 - \bar{\theta}_k B)$ can be "inverted" to the other side if $|\bar{\theta}_k| < 1$

Therefore the model is **invertible** if all $|\bar{\theta}_k| < 1 \Leftrightarrow$ all $\frac{1}{|\bar{\theta}_k|} > 1 \Leftrightarrow$ all roots lie outside the unit circle

Autoregressive Model

A stochastic process $\{Y_t\}$ follows an $AR(p)$ model if it satisfies

$Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + Z_t$, or $\phi(B)Y_t = Z_t$, where $Z_t \sim WN(0, \sigma^2)$ and $\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$ is the characteristic polynomial.

Stationarity

Lemma: If $Y_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$, $Z_t \sim WN(0, \sigma^2)$ and $\sum_{i=0}^{\infty} |\psi_i| < \infty$, then $\{Y_t\}$ is a stationary process.

Using the properties of white noise, i.e. $E(Z_t) = 0$, $\text{Cov}(Z_t, Z_k) = \sigma^2 \mathbb{1}_{\{t=k\}}$, we have

- $E(Y_t) = 0$
 - $\text{Var}(Y_t) = \sigma^2 \sum_{i=0}^{\infty} \psi_i^2 \leq \sigma^2 \left(\sum_{i=0}^{\infty} |\psi_i| \right)^2 < \infty$
 - $\text{Cov}(Y_t, Y_{t+k}) = \sigma^2 \sum_{i=0}^{\infty} \psi_i \psi_{i+k}$ is independent of t
- $\Rightarrow \{Y_t\}$ is stationary.

Theorem: The process $\phi(B)Y_t = Z_t$, $Z_t \sim WN(0, \sigma^2)$ is stationary iff no root of $\phi(x)$ is on the unit circle (i.e., all roots are outside the unit circle)

Asymptotic stationary

考虑AR(1)的展开形式, 我们有 $Y_t = Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \dots + \phi^{t-1} Z_1 + \phi^t Y_0$

我们观察到 Y_0 是一个固定值所以我们会发现

$$E(Y_t) = \phi^t E(Y_0)$$

$$\text{Var}(Y_t) = \sigma^2 (1 + \phi^2 + \dots + \phi^{2(t-1)}) + \phi^{2t} \text{Var}(Y_0), \text{ 他们是time dependent}$$

$$\text{However, as } t \rightarrow \infty, E(Y_t) \rightarrow 0 \quad \text{Var}(Y_t) = \frac{\sigma^2}{1-\phi^2}$$

Causality

Definition: Y_t is causal if it can be expressed as a linear combination of the past white noise $Z_t, Z_{t-1}, \dots$ i.e.

$$Y_t = \sum_{k=0}^{\infty} \psi_k Z_{t-k}, \text{ where } \sum_{k=0}^{\infty} |\psi_k| < \infty$$

Theorem: An $AR(p)$ process $\phi(B)Y_t = Z_t$, $Z_t \sim WN(0, \sigma^2)$ is causal if the roots of the characteristic polynomial $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ are outside the unit circle.

Proof:

$$\text{Factorization: } \phi(B) = (1 - \xi_1 B)(1 - \xi_2 B) \dots (1 - \xi_p B)$$

$$\text{The roots of } \phi(x) \text{ are } x = \frac{1}{\xi_1}, \dots, \frac{1}{\xi_p}.$$

$$\text{The roots are outside unit circle} \Leftrightarrow \left| \frac{1}{\xi_k} \right| > 1, \text{ for all } k = 1, \dots, p \Leftrightarrow |\xi_k| < 1, \text{ for all } k = 1, \dots, p \Leftrightarrow$$

$$(1 - \xi_k B), \text{ for all } k = 1, \dots, p, \text{ can be "inverted" to another side and represented by past noises.} \Leftrightarrow \{Y_t\} \text{ is causal.}$$

Yule-Walker: 对于一个stationary的Timeseries, 我们有 $\gamma_k = E[Y_t Y_{t-k}]$

ARMA

Definition: $\{Y_t\}$ is said to be an $ARMA(p, q)$ process if

$$\phi(B)Y_t = \theta(B)Z_t, \quad Z_t \sim WN(0, \sigma^2),$$

where $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ $\theta(B) = 1 - \theta_1 B - \dots - \theta_q B^q$ are the characteristic polynomials with NO common roots.

Property:

Stationary solution exists if $\Rightarrow$ All roots of $\phi(B)$ are **not on** the unit circle

Causal if $\Rightarrow$ All roots of $\phi(B)$ are **outside** the unit circle $\Rightarrow Y_t = \phi^{-1}(B)\theta(B)Z_t = \sum_{k \geq 0} \psi_k Z_{t-k}$, with

$\sum_{k \geq 0} |\psi_k| < \infty$. **Invertible** if $\Rightarrow$ All roots of $\theta(B)$ are **outside** the unit circle $\Rightarrow$

$$Z_t = \theta^{-1}(B)\phi(B)Y_t = \sum_{k \geq 0} \pi_k Y_{t-k}, \text{ with } \sum_{k \geq 0} |\pi_k| < \infty.$$

ARIMA

Noticed that for $ARIMA(p, d, q)$ model, if $d \geq 1$, the root of $(1 - B)^d$ is 1, therefore it is **non-stationary and non-causal**

But it is **invertible** if the roots of $\theta(B)$ are outside the unit circle.

Seasonal ARIMA

$\{Y_t\}$ follows an $SARIMA(p, d, q) \times (P, D, Q)_s$ model if

$$\phi(B)\Phi_P(B^s)(1 - B)^d(1 - B^s)^DY_t = \theta(B)\Theta_Q(B^s)Z_t$$

where

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$$

$$\theta(B) = 1 - \theta_1 B - \dots - \theta_q B^q$$

$$\Phi_P(B^s) = 1 - \Phi_1 B^s - \dots - \Phi_P B^{sP}$$

$$\Theta_Q(B^s) = 1 - \Theta_1 B^s - \dots - \Theta_Q B^{sQ}$$